Electronic Circuits & Applications

Unit No.: 2: MOSFET : Amplifier & Oscillator

Syllabus: Small-Signal Equivalent Circuit, CS Configuration analysis, common drain (source-follower) amplifier. Frequency response of CS amplifier.
Feedback amplifiers: Properties of negative feedback, Four types of amplifiers and feedback topologies, Examples of voltage series and Current series feedback amplifiers and their analysis.
Oscillators: Barkhausen criterion, Types of Oscillators, RC Phase Shift oscillator, LC Oscillator Hartley & Colpitts.

Mapping of Course Outcomes for Unit II:
CO2: Use small signal model of MOSFET to study its application as amplifier and oscillator

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2.1: Common Source Amplifier:

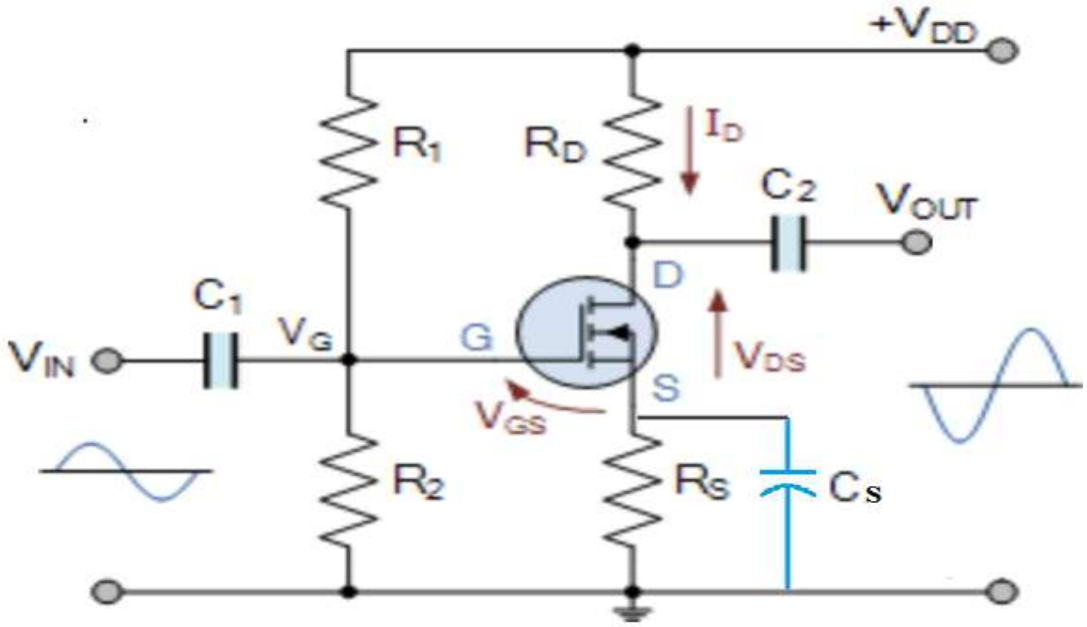


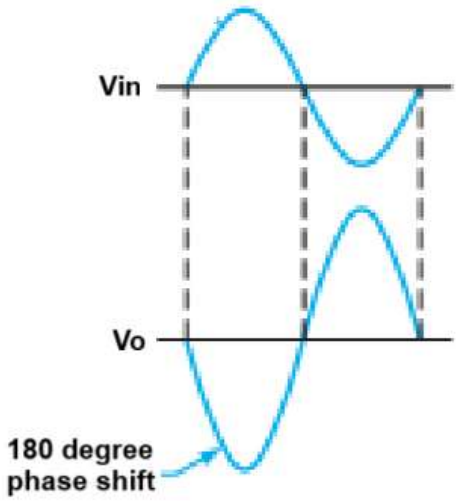
Fig.1: Common source (CS) amplifier

Working:
1) Voltage divider biasing circuit consists of 4 resistors: R1, R2, R_D & R_S.
2) Resistors R1& R2 form a voltage divider network and decide the position of Q-point on load line.
3) Resistor R_D limits the drain current and also controls the voltage gain.
4) R_S is a stabilization resistance and also controls the voltage gain.
5) C1: It is an input **coupling capacitor**, blocking DC and passing AC signals to input gate.
6) C2: It is an output **coupling capacitor**. At output drain terminal, there is mixing of both ac & dc components in amplified output. So C2 blocking DC and passing amplified pure AC output signals.
7) Cs: It is a **bypass capacitor**. When it is present and has a low impedance at the signal

frequency, it shorts the source resistor (R_S) for AC signals, increasing the AC voltage gain.

Without C_S the AC signal is fed back to the source, creating negative feedback that reduces gain but improves stability and bandwidth

8) ***Output voltage of C-S amplifier is out of phase by 180 degree.***



AC equivalent circuit of Common Source Amplifier:

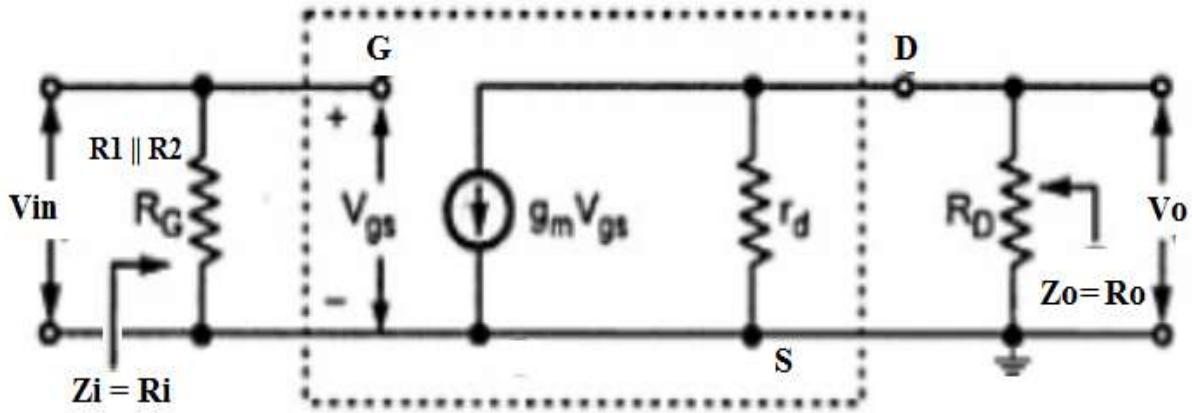


Fig.2: AC equivalent circuit of Common Source Amplifier

Mathematical formulae:

1) Transconductance: gm

It is a ratio of drain current to gate-source voltage.

$$gm = id / Vgs$$

$$gm = 2K\,(V_{GSQ} - V_{GST})$$

But $I_{DQ} = K\,(V_{GSQ} - V_{GST})^2$

$$g_m = 2 * \sqrt{K\, I_{DQ}}$$

2) Dynamic output resistance: r_o

It is ratio of change in drain voltage to change in drain current at constant gate voltage.

$$r_o = | \delta V_{DS} / \delta I_D | \quad \ldots\ldots \text{at constant } V_{GS}$$

$$\mathbf{r_o = 1 / (\lambda . I_D)}$$

3) Input impedance / Input resistance: Zi or Ri

$$Zi = Ri = R_G$$

$$\mathbf{Zi = Ri = R1 \,||\, R2}$$

4) Output impedance / Output resistance: Zo or Ro

$$\mathbf{Zo = Ro = ro \,||\, R_D}$$

If $ro >> R_D$, then

$$\mathbf{Zi = Ri \sim R_D}$$

5) Voltage gain: Av

It is ratio of output voltage to input voltage of an amplifier.

$$Av = - gm . Zo$$

$$\mathbf{Av = - gm . (ro \,||\, R_D)}$$

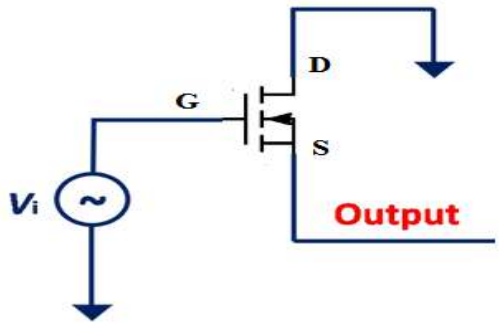
If $ro >> R_D$, then

$$\mathbf{Av = - gm . R_D}$$

2.2: MOSFET- Source Follower (Common Drain Amplifier):

In the Common Drain Amplifier configuration, the drain terminal is at AC ground.

The input is applied between the gate and drain terminals, while the output is measured between the source and drain terminal. Since the drain terminal is common between the input and output side, it is known as **Common Drain Amplifier.**



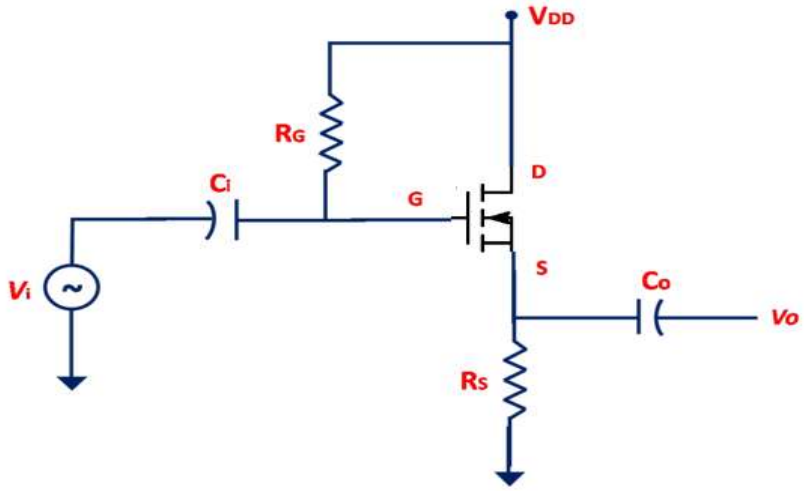


Fig.3: Common Drain Amplifier / Source follower

The source follower is a circuit with a **voltage gain slightly less than or equal to 1.00** and with a high input impedance and a low output impedance.

Hence i/p =~~ o/p

The Common Drain Amplifier has

1) High Input Impedance
2) Low Output Impedance
3) Sub-unity voltage gain

Since the output at the source terminal is following the input signal, it is also known as **Source Follower**.

2.3: Frequency response for MOSFET (C-S) amplifier

Frequency response of C-S amplifier is shown below:

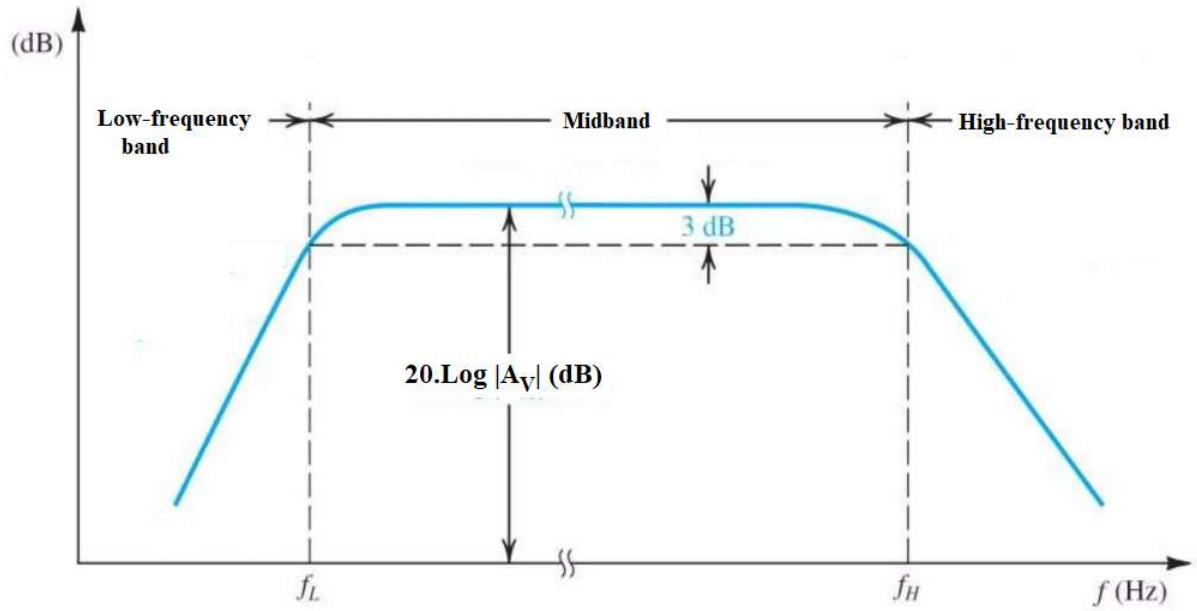


Fig.4: Frequency response of C-S amplifier

In low frequency band, gain of amplifier is low. This gain is low due to the effect of coupling & bypass capacitors. Here it is assumed that their impedances are small enough to act as short circuits.

In mid-band, the amplifier gain is maximum and remains constant w.r.t. frequency. In this mid-band, all capacitances effects are neglected.

In high frequency band, gain of amplifier is low. This gain is low due to the effects of internal capacitances.

Cut-off frequencies:

i) Lower cut-off frequency (F_L):

It is a lower frequency at which maximum gain drops by 3dB is known as Lower cut-off frequency (**F_L**).

ii) Higher cut-off frequency (F_H):

It is a lower frequency at which maximum gain drops by 3dB is known as Higher cut-off frequency (F_H).

Hence bandwidth is given as:

$$BW = F_H - F_L$$

2.4: Comparison:

Common source amplifier & Common drain amplifier:

Sr. No.	Parameter / factor	Common source amplifier	Common drain amplifier
1	***Common terminal:***	Common terminal is source w.r.t. input & output.	Common terminal is drain w.r.t. input & output.
2	***o/p voltage nature***	o/p voltage is ***out of phase*** by 180 degree with i/p	o/p voltage is ***in phase*** (zero degree) with i/p
3	***Voltage gain***	Voltage gain is maximum (high)	Voltage gain is unity
4	***Input impedance***	Input impedance is high	Input impedance is very high
5	***Output impedance***	Output impedance is low	Output impedance is very low

2.5: Feedback amplifiers: (-ve):

2.5.1: Introduction:

Feedback is a process where a portion of o/p is returned to i/p to form a part of system excitation.

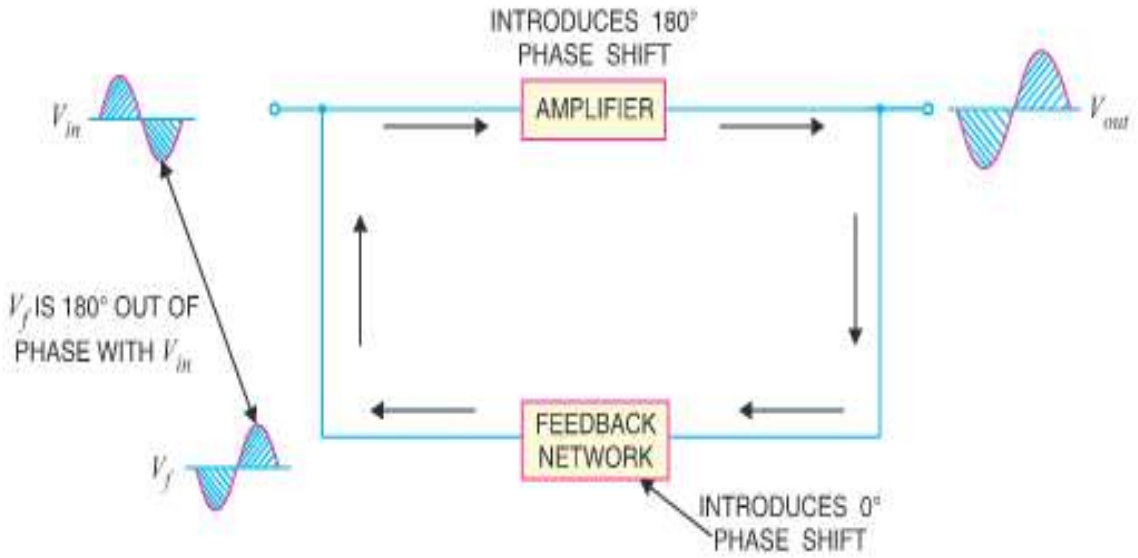
Feedback amplifier contains two components namely feedback circuit and amplifier circuit. Feedback circuit is essentially a potential divider consisting of resistances R1& R2. The purpose of feedback circuit is to return a fraction of the output voltage to the input of the amplifier circuit.

Both negative feedback and positive feedback are used in amplifier circuits.
Negative feedback returns part of the output to oppose the input, whereas in positive feedback the feedback signal aids the input signal.

For negative feedback: βA > 0;
For positive feedback: βA < 0

Negative feedback:

When the feedback energy (voltage or current) is out of phase with the input signal and thus opposes it, it is called negative feedback. This is illustrated in Fig. The amplifier introduces a phase shift of 180° into the circuit while the feedback network is so designed that it introduces no phase shift (i.e., 0° phase shift). The result is that the feedback voltage V_f is 180° out of phase with the input signal V_{in}.



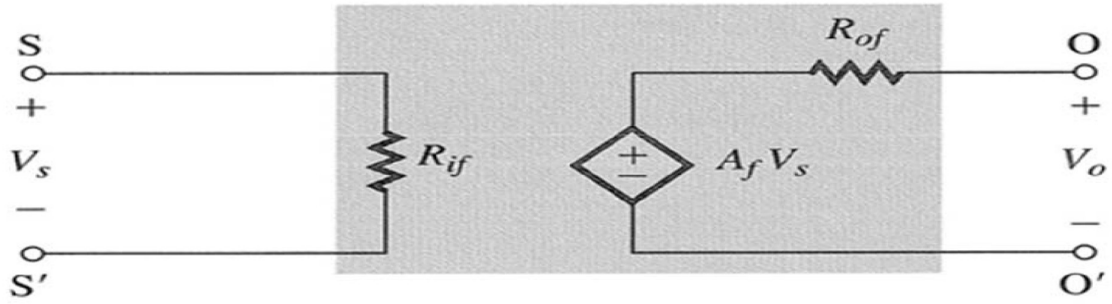
Effects of Negative feedback:

1) Voltage gain reduces (output voltage decreases)
2) Nonlinear distortion reduces
3) Input impedance / resistance increases
4) Output impedance / resistance decreases
5) Bandwidth increases
6) Effect of noise reduces
7) Gain stability increases

2.5.2: Basic Amplifiers in feedback network:

1) Voltage amplifier:

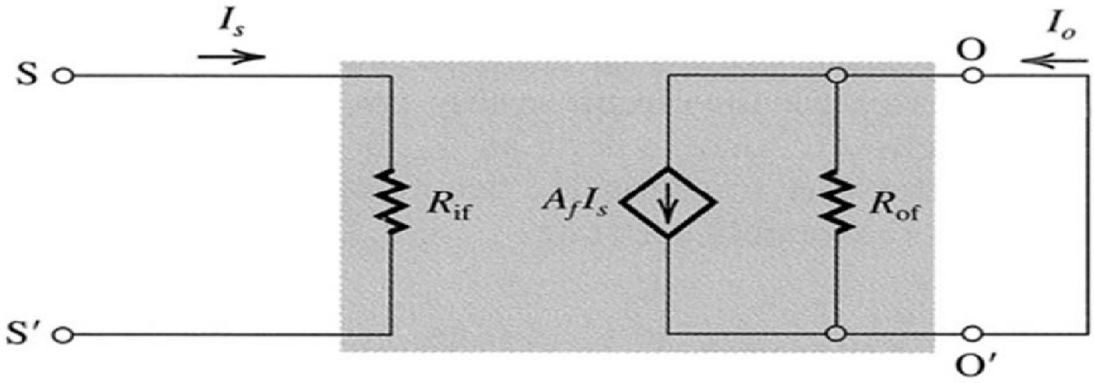
Consider a *thevenin's equivalent* of a voltage amplifier



- `Ri >> Rs` `ideally Ri = ∞`
- R_L >> Ro ideally Ro = 0
- It is used in *Voltage series feedback amplifier*

2) Current amplifier:

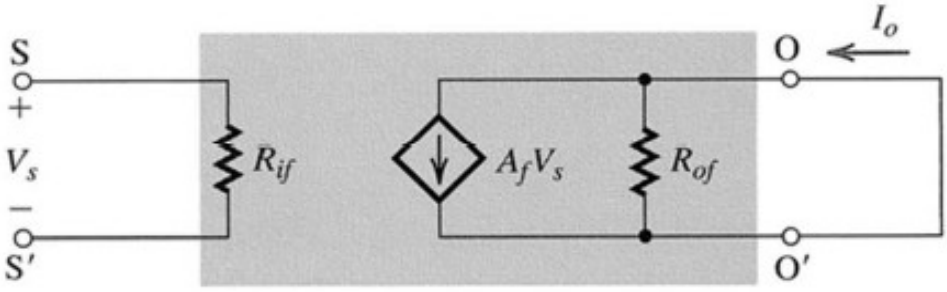
Consider a *Norton's equivalent* of a current amplifier



- Ri << Rs ideally Ri = 0
- R_L `<< Ro` `ideally Ro = ∞`
- It is used in *Current shunt feedback amplifier*

3) Transconductance amplifier:

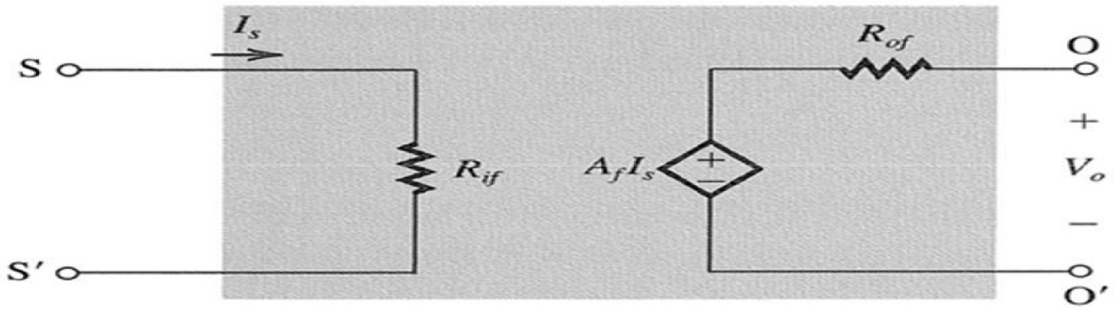
Consider a transconductance amplifier with a *thevenin's equivalent* in its input and *Norton's equivalent* in its output.



- Ri >> Rs ideally Ri = ∞
- R_L << Ro ideally Ro = ∞
- It is used in *Current series feedback amplifier*

4) Trans-resistance amplifier:

Consider a transconductance amplifier with a *Norton's equivalent* in its input and *thevenin's equivalent* in its output.

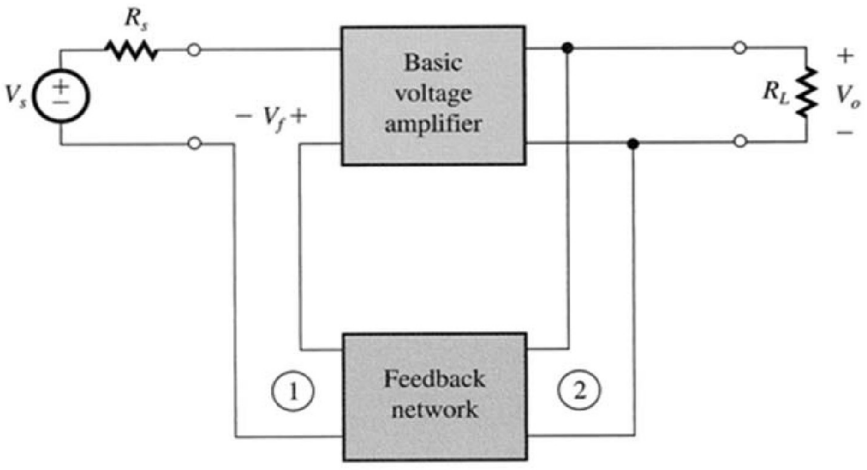


- Ri << Rs ideally Ri = 0
- R_L >> Ro ideally Ro = 0
- It is used in *Voltage shunt feedback amplifier*

2.5.3: Different topologies: (types of feedback amplifiers)

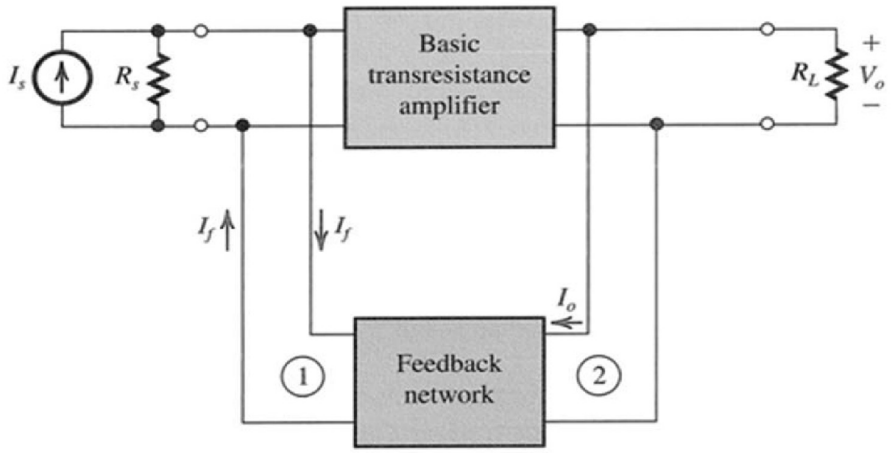
There are 4 types of topologies of negative feedback

1) Voltage series feedback amplifier:



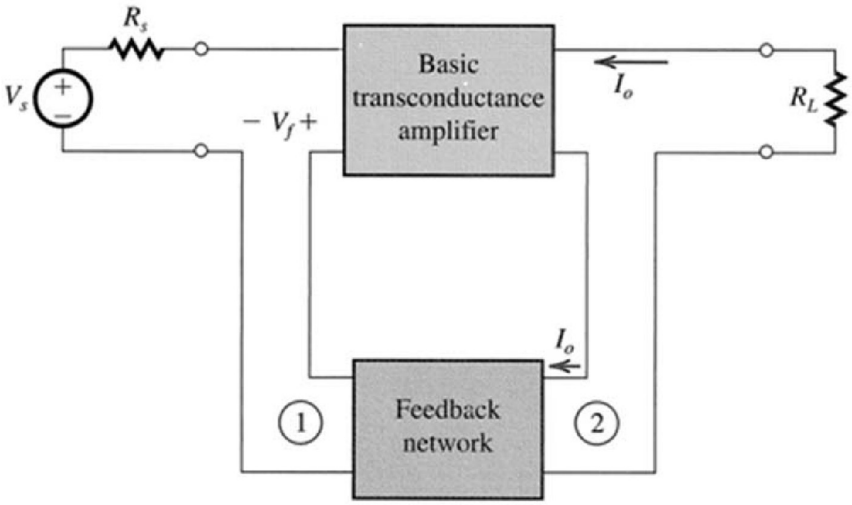
i) Basic amplifier : *Voltage amplifier*
ii) Sampling type (signal) : *Voltage*
iii) Type of mixing : *Series*
iv) Input resistance with feedback (Rif) : *Increases*
v) Output resistance (Rof) : *Decreases*

2) Voltage shunt feedback amplifier:



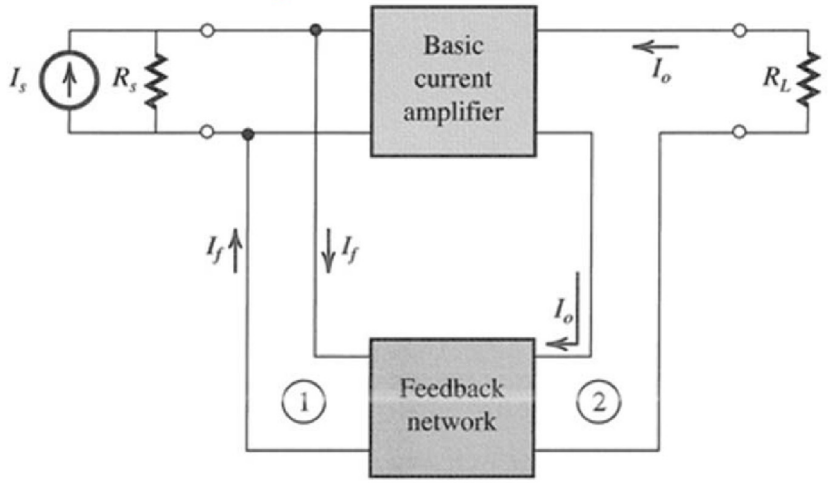
i) Basic amplifier : *trans-resistance amplifier*
ii) Sampling type (signal) : *Voltage*
iii) Type of mixing : *Shunt*
iv) Input resistance with feedback (Rif) : *Decreases*
v) Output resistance (Rof) : *Decreases*

3) Current series feedback amplifier:



i) Basic amplifier : *transconductance amplifier*
ii) Sampling type (signal) : *Current*
iii) Type of mixing : *Series*
iv) Input resistance with feedback (Rif) : *Increases*
v) Output resistance (Rof) : *Increases*

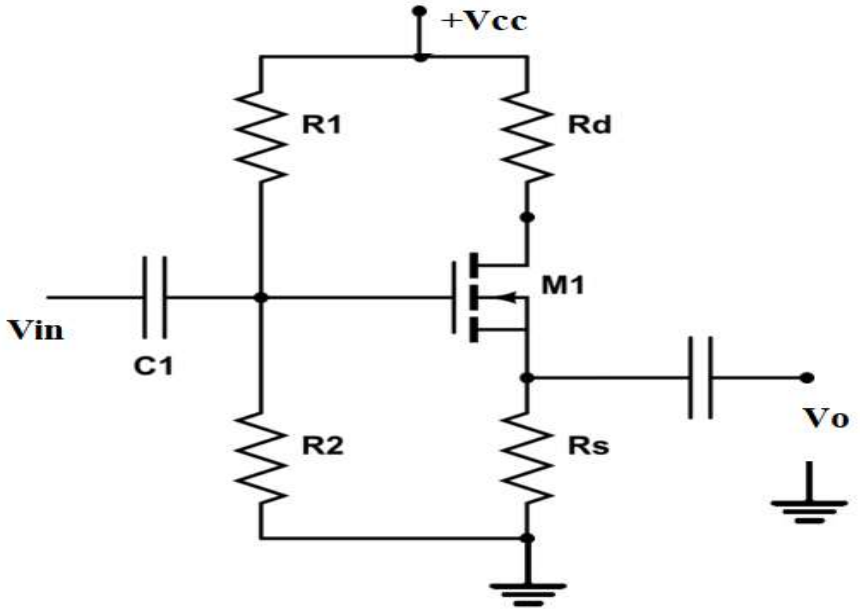
4) Current shunt feedback amplifier:



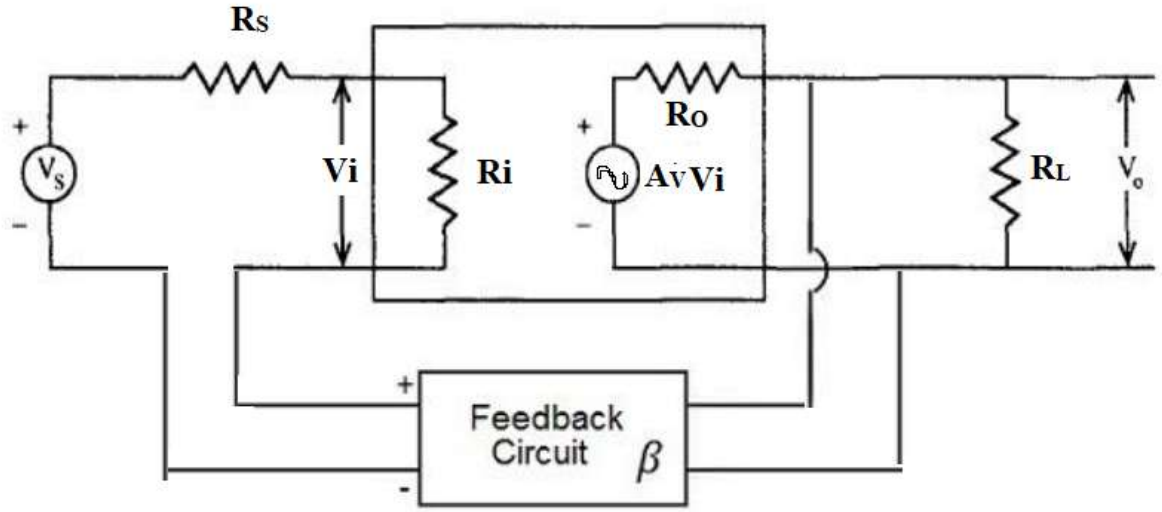
i) Basic amplifier : *Current amplifier*
ii) Sampling type (signal) : *Current*
iii) Type of mixing : *Shunt*
iv) Input resistance with feedback (Rif) : *Decreases*
v) Output resistance (Rof) : *Increases*

2.6: Voltage series feedback amplifier:

Circuit diagram of voltage series feedback amplifier is shown below:



Its ac model is shown below:



Mathematical formulae:

1) Input resistance with feedback: Rif:

$$Rif = Ri\ (1+Av * \beta)$$

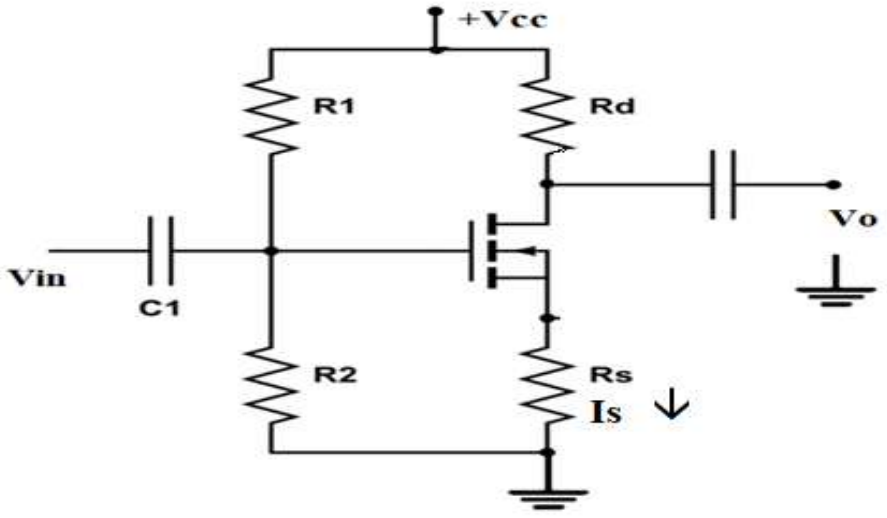
2) Output resistance with feedback: Rof:

$$Rof = Ro\ /\ (1+Av * \beta)$$

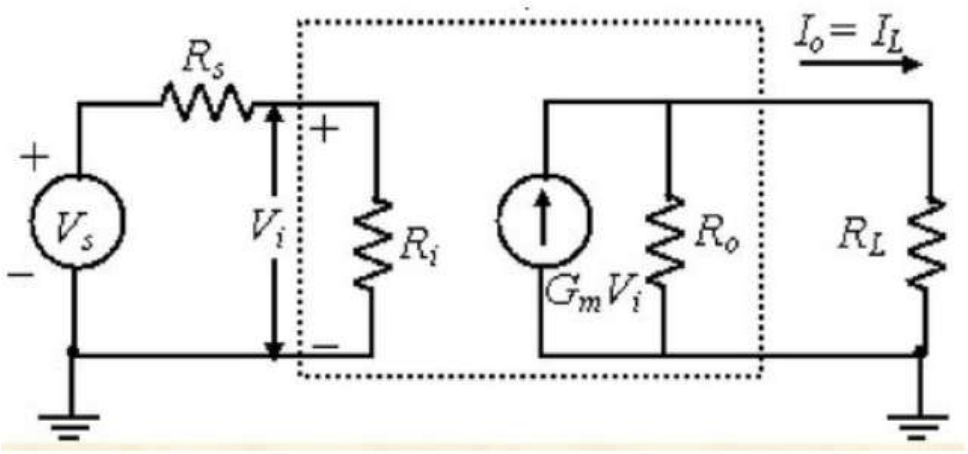
3) Voltage gain: Avf:

$$Avf = Av\ /\ (1+Av * \beta)$$

2.7: Current series feedback amplifier:



Its ac model is shown below:



Mathematical formulae:

1) Input resistance with feedback: Rif:

$$Rif = Ri * (1+ \beta*Gm)$$

2) Output resistance with feedback: Rof:

$$Rof = Ro * (1+ \beta*Gm)$$

3) Voltage gain: Avf:

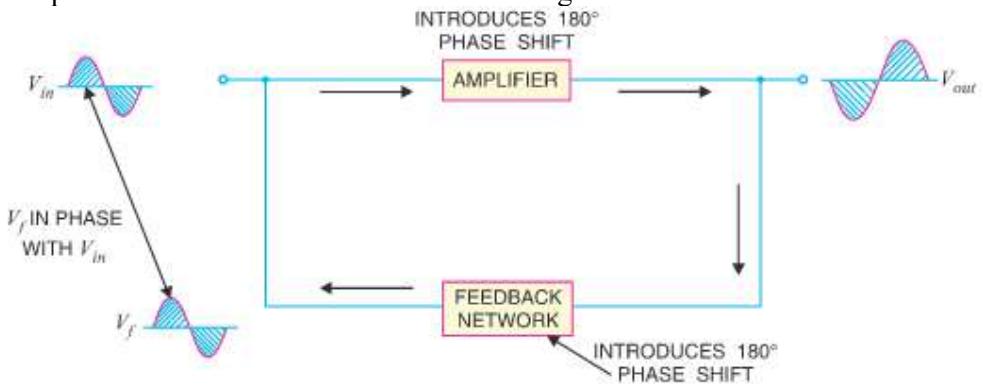
$$Avf = Av / (1+Av * \beta)$$

Comparison of feedback amplifiers:

Sr. No.	Parameters	Voltage-Series	Current-Series	Voltage-Shunt	Current-Shunt
1	Voltage Gain: Av	Decreases	Decreases	Decreases	Decreases
2	Bandwidth: BW	Increases	Increases	Increases	Increases
3	Input resistance: Ri	**Increases**	**Increases**	Decreases	Decreases
4	Output resistance: Ro	**Decreases**	Increases	**Decreases**	Increases
5	Harmonic distortion	Decreases	Decreases	Decreases	Decreases
6	Noise	Decreases	Decreases	Decreases	Decreases

2.8: Positive feedback:

When the feedback energy (voltage or current) is in phase with the inputsignal and thus aids it, it is called positive feedback . This is illustrated in Fig.



Both amplifier and feedback network introduce a phase shift of 180°. The result is a 360° phase shift around the loop, causing the feedback voltage Vf to be in phase with the input signal Vin.

The positive feedback increases the gain of the amplifier. However, it has the disadvantages of increased distortion and instability. Therefore, positive feedback is seldom employed in amplifiers. One important use of positive feedback is in oscillators

2.8.1: The Barkhausen Criterion:

Oscillations will not be sustained if, at the oscillator frequency, the magnitude of the product of the transfer gain of the amplifier and the magnitude of the feedback factor of the feedback network (the magnitude of the loop gain) are less than unity.

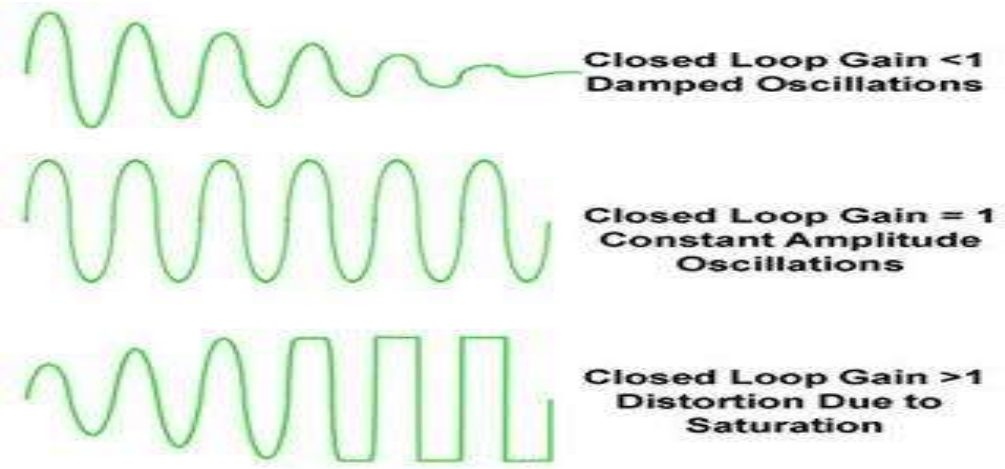
· The condition of *unity loop gain* $-A\beta = 1$ is called the *Barkhausen criterion*. This condition implies, of course, both that $|A\beta| = 1$ and that the phase of $-A\beta$ is zero. The above principles are consistent with the feedback formula $A_f = A/(1 + \beta A)$. For if $-\beta A = 1$, then $A_f \to \infty$, which may be interpreted to mean that there exists an output voltage even in the absence of an externally applied signal voltage.

An oscillator must have an amplifier, a positive feedback loop and some method of controlling the frequency of oscillation.

In RF sine wave oscillators, the frequency may be controlled by an LC tuned circuit, but as well as controlling the frequency of oscillation, there must also be some means, such as negative feedback, of stabilizing the amplitude of the signal produced.

Without this stabilization the oscillations would either die away and stop (damped oscillation) or rapidly increase in amplitude until the amplifier produces severe distortion due to the transistors within the amplifier becoming "saturated" as shown in Fig.

To produce a constant amplitude output the gain of the amplifier is automatically controlled during oscillation.

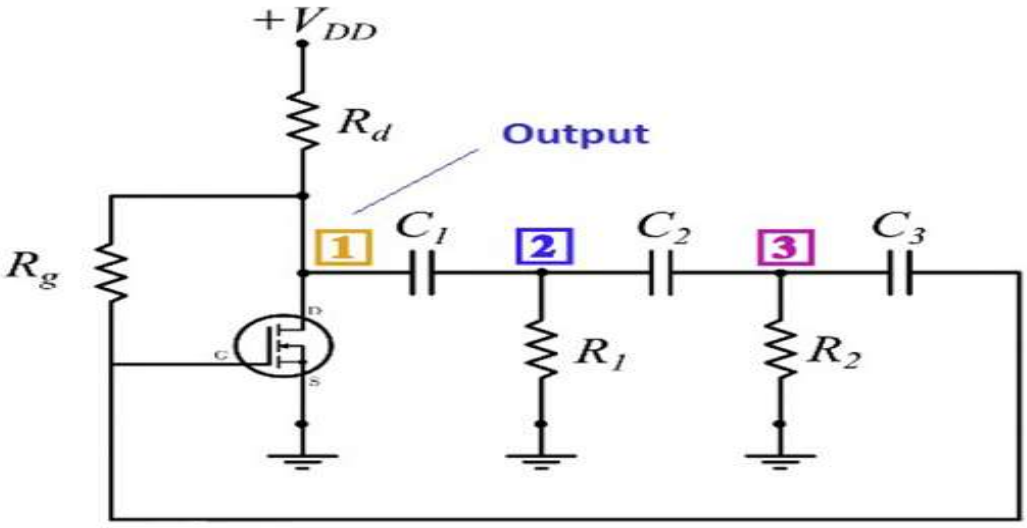


2.8.2: R-C Phase Shift Oscillator:

The circuit consists of a common source MOSFET amplifier followed by a three section R-C phase shift network. The amplifier stage is self-biased with a capacitor bypassed source resistor R_s and a drain bias resistance R_D. The output of the last section is supplied back to the gate. If the loading of the phase-shift network on the amplifier can be assumed to be negligible, a phase shift of 180° between the amplified output voltage V_{out} and the input voltage V_{in} at the gate is produced by the amplifier itself.

The three-section R-C phase shift network produces an additional phase shift, which is a function of frequency and equals 180° at some frequency of operation. At this frequency the total phase shift from the gate around the circuit and back to gate will be exactly zero. This particular frequency will be the one at which the circuit will oscillate provided that the magnitude of the amplification is sufficiently large. In a FET phase-shift oscillator voltage series feedback [that is, feedback voltage proportional to the output voltage V_{out} and supplied in series with the input signal at the gate is used.

The frequency of the oscillator output depends upon the values of capacitors C and resistors R used in the phase shift network. Using basic RC circuit analysis technique, it can be shown that the network phase shift is 180°



Frequency of R-C phase shift oscillator is given as:

$$f_r = \frac{1}{2\pi RC\sqrt{2N}} = \frac{1}{2\pi RC\sqrt{6}}$$

Where: f r is the Output Frequency in Hertz

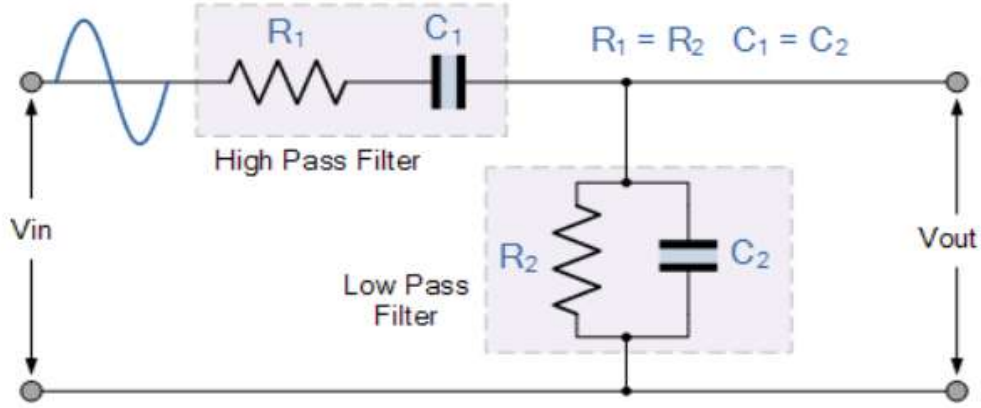
R is the Resistance in Ohms

C is the Capacitance in Farads

N is the number of RC stages. (N = 3)

2.8.3: Wein Bridge Oscillator:

The **Wien bridge oscillator** is used to find unknown values of components. In most of the cases this oscillator is used in the audios. The oscillators are designed simply, size is compressed and it has stable in frequency output. Hence the maximum output frequency of the Wien bridge oscillator is 1MHz and this frequency is from the phase shift oscillator. The total **phase shift of the oscillator** is from the 360° or 0°.



It is a two stage amplifier with **RC bridge circuit** and the circuit has the lead lag networks. The lags at the phase shift are increasing the frequency and the leads are decreasing the frequency.

In additional by adding the Wien Bridge oscillator at a particular frequency it becomes sensitive. At this frequency the Wien Bridge is balance the phase shift of 0°.

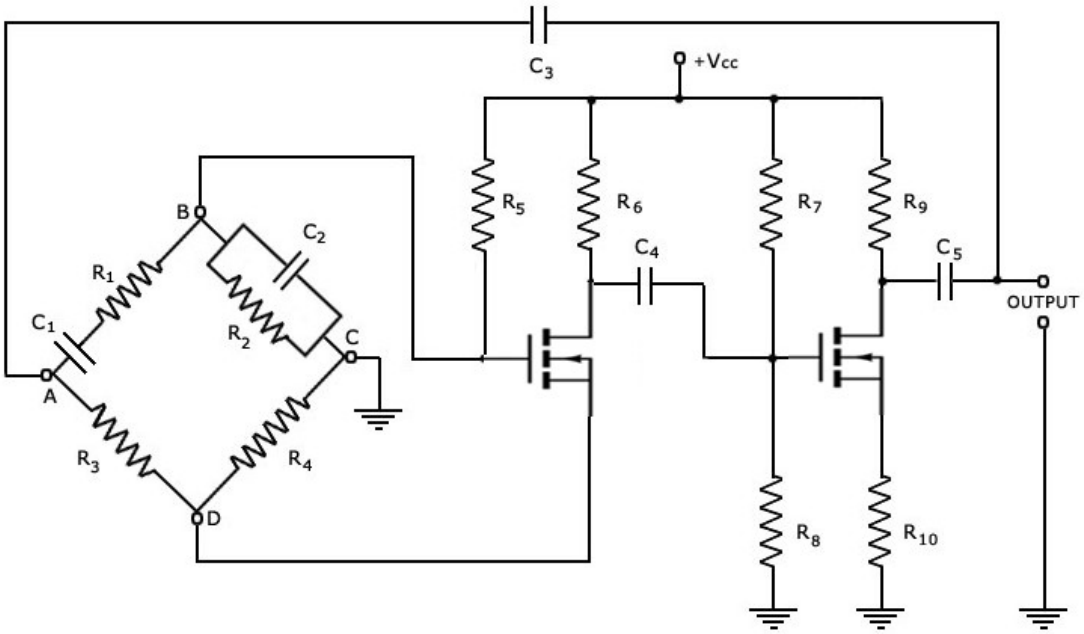
The following diagram shows the circuit diagram of the Wien bridge oscillator. The diagram shows R1 is series with the C1, R3, R4 and R2 are parallel with the C2 to from the four arms.

Wien Bridge Oscillator Frequency

$$f_r = \frac{1}{2\pi RC}$$

Where:

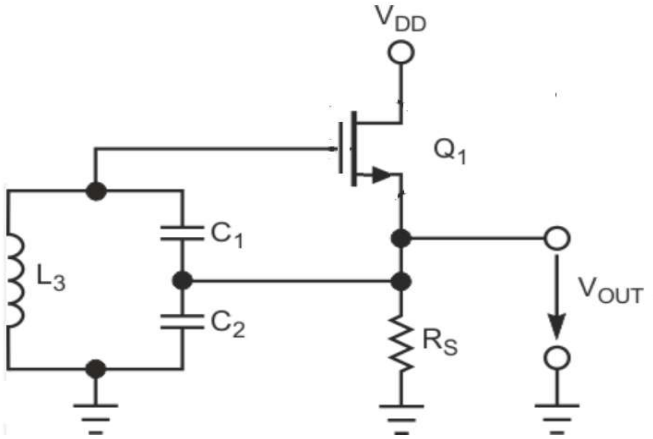
- fr is the Resonant Frequency in Hertz
- R is the Resistance in Ohms
- C is the Capacitance



The circuit is in the oscillation mode and the gate voltage of the first MOSFET is changed randomly because it is due to the difference in voltage of DC supply. The gate voltage is applied to the drain terminal of the first transistor and the phase shift is about the 180°.

The output of the first MOSFET is given to the gate terminal of the second MOSFET with the help of the capacitor C4. Further, this process is amplified and from the second MOSFET of drain terminal the phase reversed signal is collected i.e. (oscillations produced).

2.8.4: Colpitt's Oscillator:



The source terminal of the MOSFET is effectively connected to the junction of the two capacitors, C1 and C2 which are connected in series and act as a simple voltage divider. When the power supply is firstly applied, capacitors C1 and C2 charge up and then discharge through the

coil L. The oscillations across the capacitors are applied to the gate and appear in the amplified at the source output.

Resistor, Rs provides the usual stabilizing DC bias for the MOSFET in the normal manner while the additional capacitors act as a DC-blocking bypass capacitors. An inductance L3 is used in the source circuit to provide a high reactance at the frequency of oscillation, (fr) and a low resistance at DC to help start the oscillations.

The amount of feedback is determined by the ratio of C1 and C2. These two capacitances are generally “ganged” together to provide a constant amount of feedback so that as one is adjusted the other automatically follows.

The frequency of oscillations for a Colpitts oscillator is determined by the resonant frequency of the LC tank circuit and is given as:

$$f_r = \frac{1}{2\pi\sqrt{LC_T}}$$